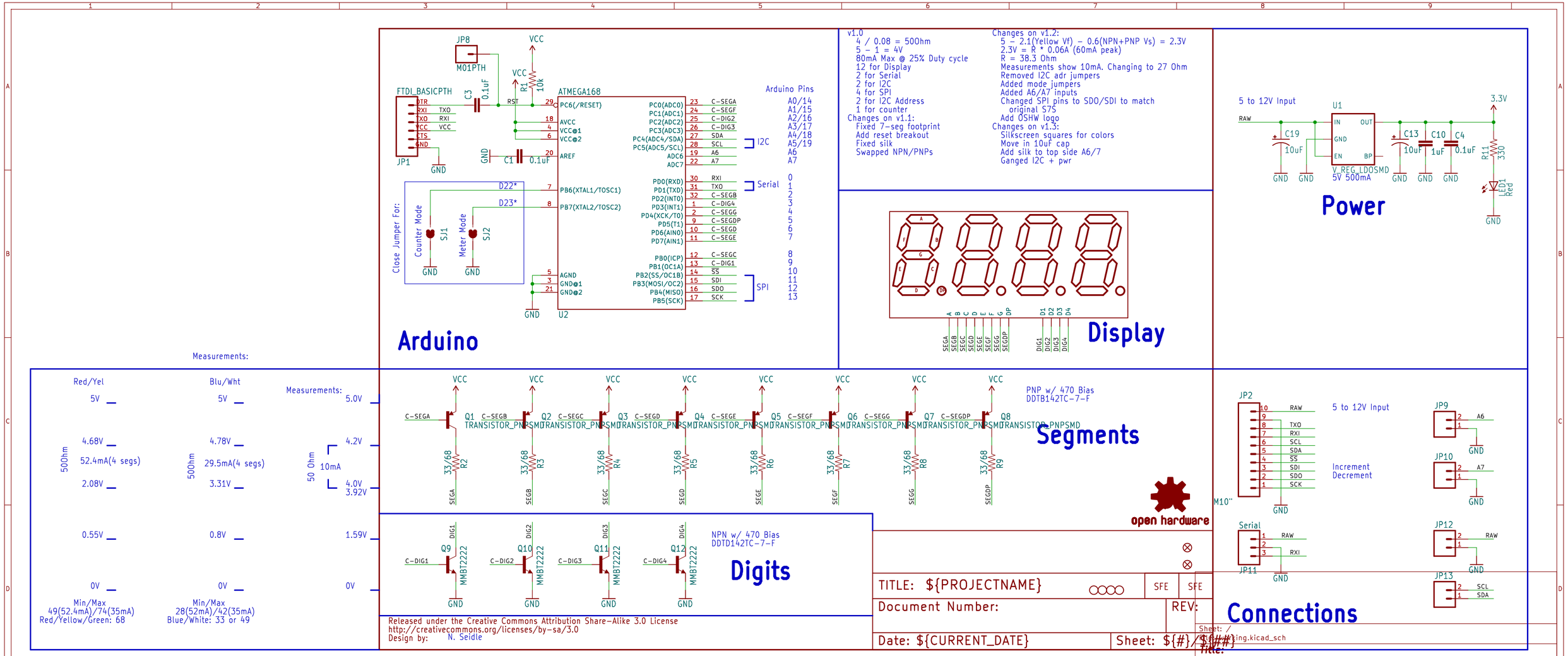
v1.0
4 / 0.08 = 50Ohm
5 - 1 = 4V
80mA Max @ 25% Duty cycle
12 for Display
2 for Serial
2 for I2C
4 for SPI
2 for I2C Address
1 for counter
Changes on v1.1:
Fixed 7-seg footprint
Add reset breakout
Fixed silk
Swapped NPN/PNPs
Changes on v1.2:
5 - 2.1(Yellow Vf) - 0.6(NPN+PNP Vs) = 2.3V
2.3V = R * 0.06A (60mA peak)
R = 38.3 Ohm
Measurements show 10mA. Changing to 27 Ohm
Removed I2C adr jumpers
Added mode jumpers
Added A6/A7 inputs
Changed SPI pins to SDO/SDI to match original S7S
Add OSHW logo
Changes on v1.3:
Silkscreen squares for colors
Move in 10uF cap
Add silk to top side A6/7
Ganged I2C + pwr
Arduino
Display
Power
Segments
Digits
Connections
Measurements:
Red/Yel
Blu/Wht
5V
4.68V
52.4mA(4 segs)
2.08V
0.55V
0V
4.78V
29.5mA(4 segs)
3.31V
0.8V
0V
500hm
50 Ohm
10mA
5.0V
4.2V
4.0V
3.92V
1.59V
0V
Min/Max
49(52.4mA)/74(35mA)
Red/Yellow/Green: 68
Min/Max
28(52mA)/42(35mA)
Blue/White: 33 or 49
PNP w/ 470 Bias
DDTB142TC-7-F
NPN w/ 470 Bias
DDTD142TC-7-F
5 to 12V Input
Increment
Decrement
open hardware
TITLE: ${PROJECTNAME}
Document Number:
REV:
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}
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Design by: N. Seidle
Sheet: /
File: Counting.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. eeschema 7.0.6
Id: 1/1